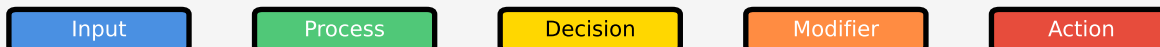
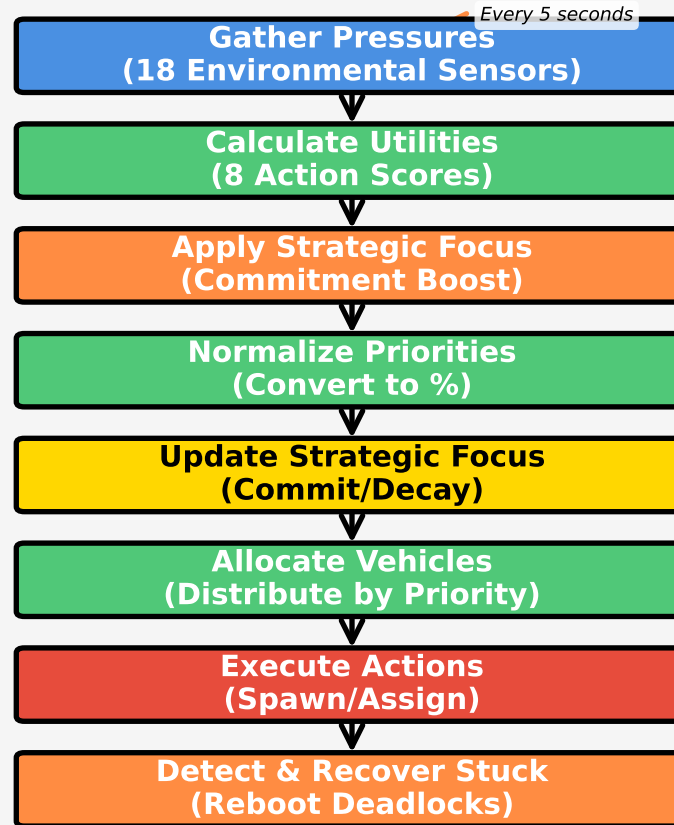


AI Decision-Making Pipeline

Planet Conquest - Complete Flow



Stage 1: Gather Environmental Pressures

18 Real-Time Sensors • Updated Every 5 Seconds

Economic Sensors

ProximalResources (unclaimed within 3000u)

OwnedResources (count)

Money (current balance)

Military Sensors

VehiclesOwned (count)

VehiclesInCombat (active fighters)

ProximalEnemyVehicles (within 3000u)

Territory Sensors

CitiesOwned (count)

FactoriesOwned (count)

TurretsOwned (count)

Threat Sensors

ProximalEnemyCities (within 6000u)

IncomingThreats (5-sec prediction)

CitiesUnderActiveAttack (HP lost)

Status Sensors

LowestCityHealthPercent (weakest city)

StuckVehicles (position tracking)

TotalIncome (per 5-sec tick)

Strategic Sensors

ControlPercentage (territory %)

EnemyStrength (relative power)

ExpansionOpportunities (available cities)

Incoming Threats:
Velocity-based
5-sec lookahead

Stuck Vehicles:
Excludes active
captures/attacks

Cities Under Attack:
HP delta tracking
between updates

18 Pressure Values → Calculate Utilities

Key: Pressures are normalized observations (0.0-1.0 or counts). They feed into utility calculations.

Stage 2: Calculate Utility Scores

8 Actions Evaluated • Personality-Modified

Capture Resources

Base: $0.7 + \text{Greed} \times 0.3$
+Proximity, +Safety, +Money
Saturates at 10 resources

Base: $0.5 + \text{Agg} \times 0.3 + \text{Greed} \times 0.2$
+Conflict willingness

Build Vehicles

Base: 0.5
+Safety bonus, 1k–3k threshold
Scales with money & threats

Base: $0.2 + \text{Def} \times 0.3$
REQUIRES threats, 2.5k–6k
Situational defense

Build Factories

Base: $0.5 + \text{Greed} \times 0.4$
Income calc, ROI 2.5min
Cap at 15 factories

Assess all cities
 $\text{Confidence} \times 0.7 + \text{Value} \times 0.3$
Value = Weakness $\times$ Resources

Base: $\text{Agg} \times 0.7 + \text{Def} \times 0.3$
+Superiority awareness
Targets nearby enemies

Base: $0.6 + \text{Def} \times 0.4$
+Incoming threats $\times 1.5$
+Active attack urgency

Personality Modifiers Applied:

Overextender ($\text{Agg} \times \text{Greed} > 0.6$): $0.7 \times$ weak cities, $1.2 \times$ resources
Fortress ($\text{Agg} \times \text{Def} > 0.6$): $1.3 \times$ strong cities, $1.2 \times$ defense
Turtle ($\text{Greed} \times \text{Def} > 0.6$): $1.2 \times$ factories, $0.8 \times$ distant targets

8 Base Utilities $\times$ Personality $\rightarrow$ Focus Boost

Key: Each utility is calculated from pressures + personality traits, then modified by archetypes.

Stage 3 & 5: Strategic Focus System

Commitment & Momentum • Prevents Flip-Flopping

Stage 3: Apply Focus Boost

IF CurrentStrategicFocus exists:
 $Utility[Focus] *= (1.0 + FocusStrength \times 0.5)$
Max boost: +50% when fully committed

After normalization:
IF Best action >35% (Agg) or >25% (Others):
Set CurrentStrategicFocus
FocusStrength = 0.7
ELSE:
FocusStrength *= (1.0 - DecayRate)

Decision: Should we commit to this action?

YES: Winning Decisively

- >35% if Aggression high
- >25% otherwise
- Set focus strength 0.7

NO: Close Competition

- Decay existing focus
- Rate: 15% (low AttSpan) to 3% (high AttSpan)

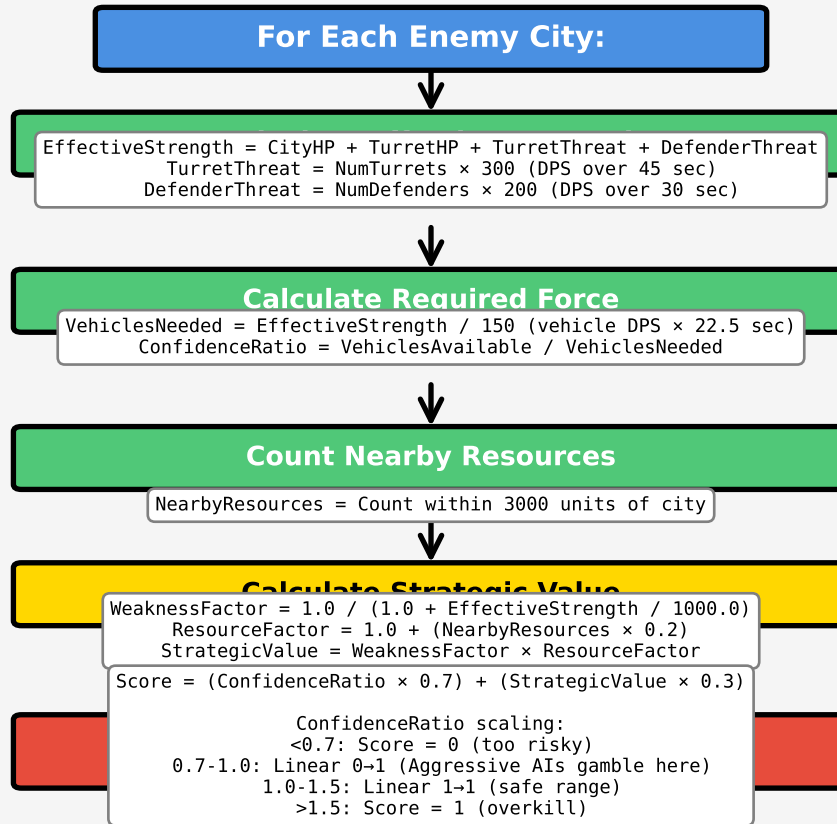
Special Hold: City Offensive

IF attacking city AND (Money > \$2000 OR under attack ourselves):
Don't abandon offensive (maintain focus)

Key: High AttentionSpan = stubborn (3% decay), Low AttentionSpan = distractible (15% decay).

City Assessment & Smart Targeting

Strategic Value = Weakness × Resources



Key: Weak cities with many resources score higher. AI considers both winnable fights AND strategic payoff.

Stage 7 & 8: Execution & Recovery

Action Execution • Stuck Vehicle Handling

Stage 7: Execute Actions

Based on normalized priorities:

1. Spawn buildings on cities
 - Vehicles, Turrets, Factories
 - Check money & limits
2. Assign vehicles to tasks
 - Capture resources
 - Attack cities/vehicles
 - Defend cities
3. Set bHasAssignment flag

Stage 8: Detect Stuck Vehicles

Position tracking between updates:

FOR each vehicle with assignment:
SKIP if TargetResource $\neq$ null
 (actively capturing)
SKIP if CurrentTarget valid
 (actively attacking)

IF moved < 100 units:
 Mark as stuck

Stuck Vehicle Recovery

Find Nearest Friendly City

Calculate Opposite Side of City

```
ToCityDir = (City - Vehicle).Normalized
Distance = |City - Vehicle|
OppositePos = City + (ToCityDir × Distance)
```

Teleport & Clear Assignment

OLD: Planet center ref
→ Sent across world

FIXED

NEW: City center ref
→ Stays in territory

Example Scenario: Aggressive AI Mid-Game

Personality: Agg=0.8, Greed=0.6, Def=0.3, Attention=0.7

Situation: 2 cities owned, 5 vehicles, \$3500, enemy city nearby (weak, 3 resources)

Pressures Gathered

ProximalEnemyCities=1, Money=3500, VehiclesOwned=5
ProximalResources=2, OwnedResources=4, IncomingThreats=2

Utilities Calculated

CaptureResources=0.85, CaptureCities=0.92 (high!)
BuildFactories=0.78, DefendCities=0.45
Other actions <0.5

Personality Applied

Overextender detected ($\text{Agg} \times \text{Greed} = 0.48 < 0.6$)
No major modifiers, base utilities stand

Focus Boosted

CurrentFocus=None (first decisive action)
No boost applied this round

Normalized

After normalization:
CaptureCities=38%, CaptureResources=28%
BuildFactories=25%, Others=9%

Focus Updated

CaptureCities > 35% (aggression threshold)
Set CurrentStrategicFocus=CaptureCities
FocusStrength=0.7

Vehicles Allocated

CaptureCities gets 38% of 5 = 2 vehicles
CaptureResources gets 28% of 5 = 1 vehicle
BuildFactories executes (has money)

Next Update

Next tick: CaptureCities gets +50% boost
Utility: $0.92 \times 1.35 = 1.24$ (massive lead)
Committed to city attack until won or desperate

AI System Summary

Planet Conquest Decision-Making Intelligence

Economic Reasoning

- Factories prioritized over turrets
- ROI-aware building (2.5min payback)
 - Income vs cost calculations
 - Smart resource saturation

Strategic Commitment

- No flip-flopping between strategies
- AttentionSpan-based persistence
- Special holds for city offensives
 - Momentum system

Smart Combat

- City strength assessment
- Confidence-based targeting
 - Overkill ratio (1.2-1.5x)
- Won't suicide into strong cities

Resource Intelligence

- Strategic value calculation
- Weakness x Resources scoring
- 3000-unit proximity awareness
 - Enemy resource denial

Threat Prediction

- 5-second velocity lookahead
 - Incoming threat tracking
- Preemptive defense (1.5x mult)
 - Active attack urgency

Personality Archetypes

- Overextender: Rushes weak targets
- Fortress: Waits for strong force
- Turtle: Builds economy first
 - Emergent behaviors

System Statistics

18 Environmental Sensors • 8 Action Types • 4 Personality Traits
3 Personality Archetypes • 5-Second Update Cycle
Strategic Focus Boost: 0-50% • Commitment Thresholds: 25-35%
Stuck Vehicle Detection • Position Tracking • Smart Recovery